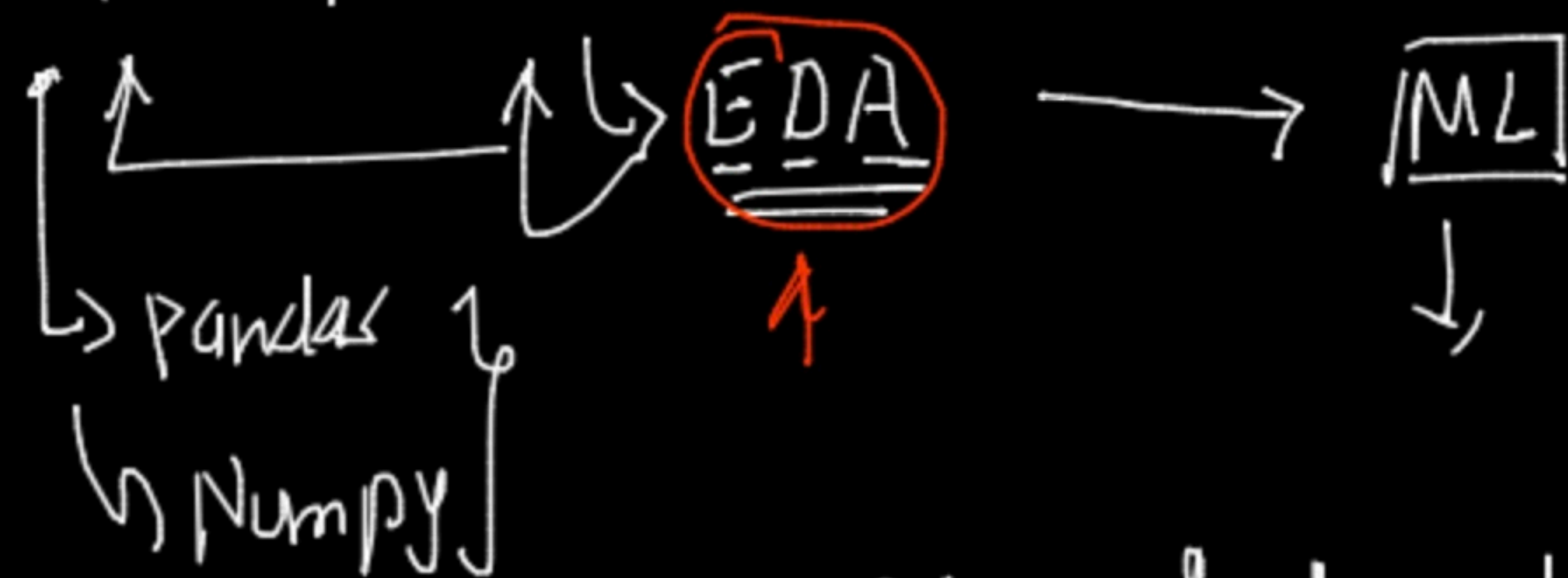


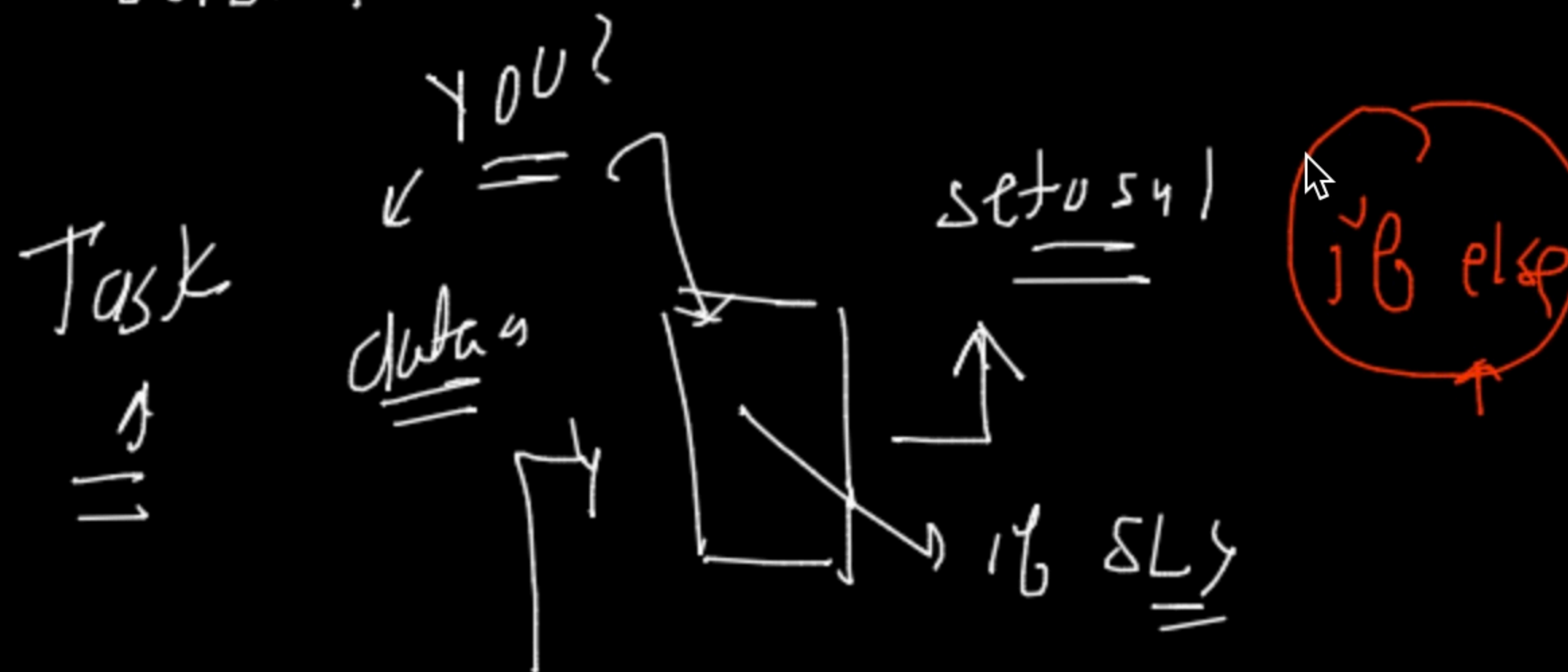
Python + Maths



eg. iris dataset

SL	PL	PL0	PL1
5.1	-	-	-
4.9	-	-	-
5.0	-	-	-
5.4	-	-	-
5.5	-	-	-
5.2	-	-	-
5.7	-	-	-
6.3	-	-	-
6.4	-	-	-

categories { setosa, versicol, virginica }



[SL, PL, PL0, PL1]

→ Problem → ?

Data →

Banking

Historical

status → yes / No

classification problem?

	C0	C1	C2	C3	Y
	credit_score	income	# of active	# of closed	Status
(	-	-	-	-	Yes ✓
(	-	-	-	-	Yes ✓
-	-	-	-	-	No
					⋮

C_{new} → (C₀ C₁ C₂ C₃) → Yes / No

solution → Rule group
→ Analysis

Historical

$D(x, y)$

(data)

ML

Supervised ML

Unsupervised ML

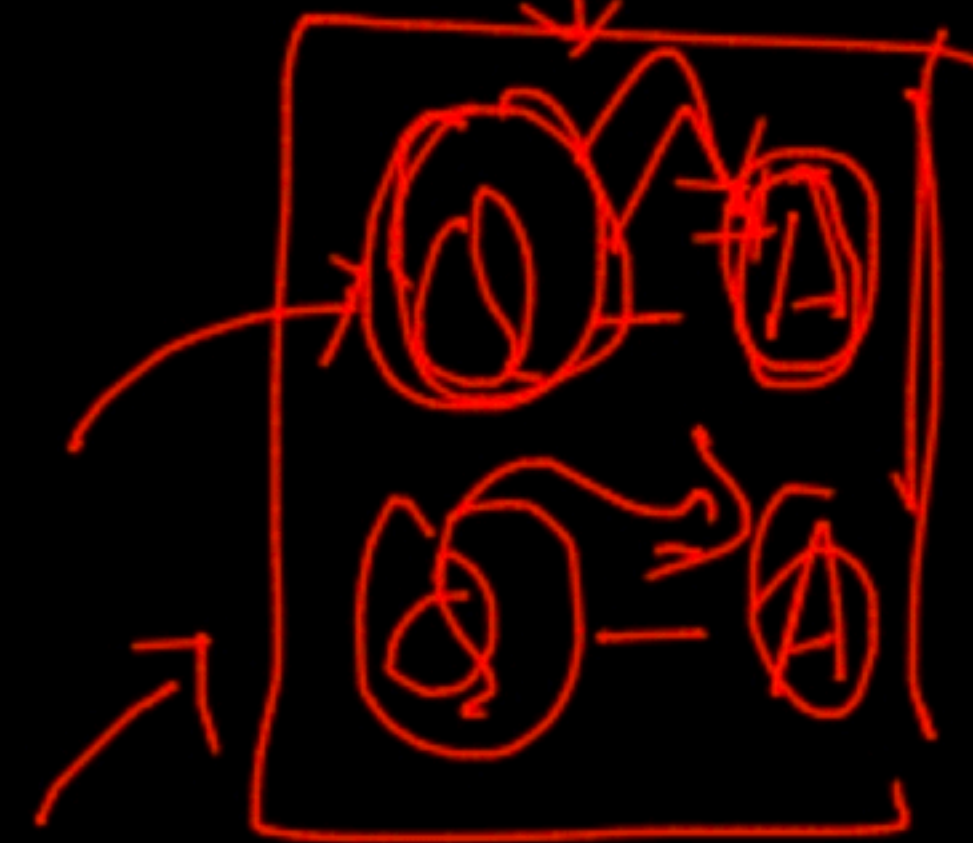
(i) Classification

Regression

Exam

(data)

Buy Book



Score ↑



New

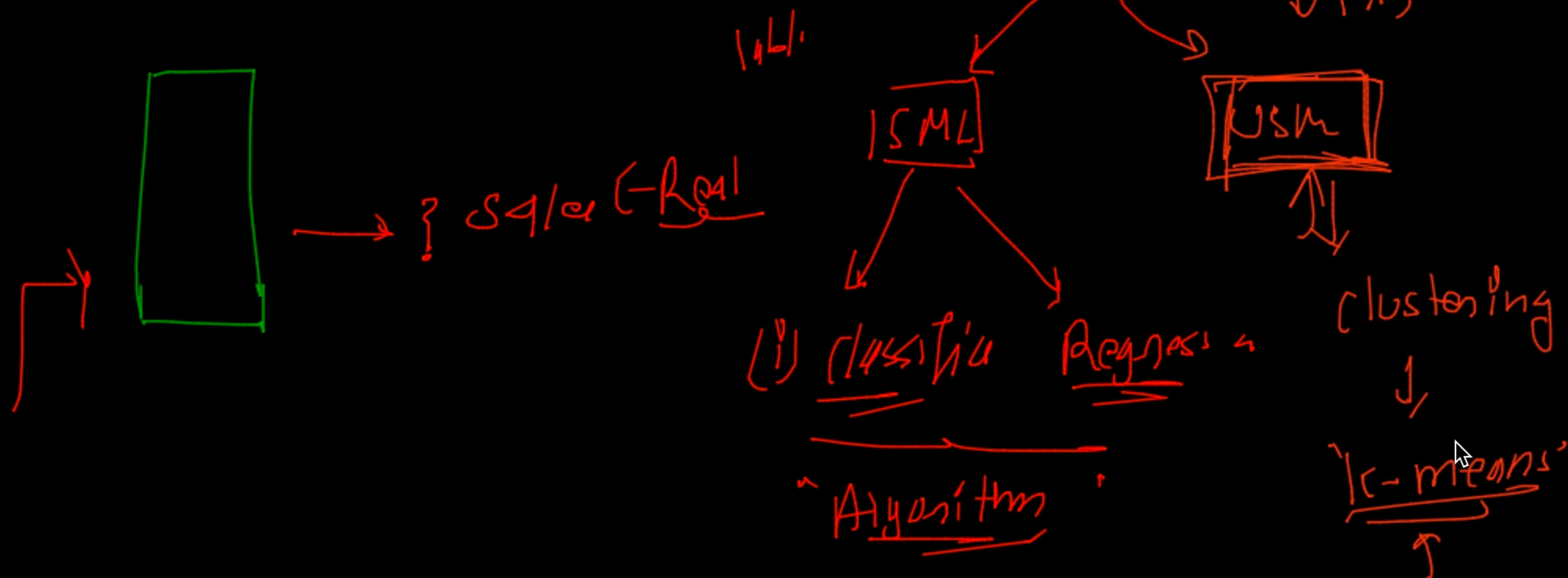
Retail

data

historical
data

X			Y → Real
discount	B/w	is weekend	sale
0.10	T	NO	350
⋮	⋮	⋮	⋮
[0,2	0	yes	

Regression problem



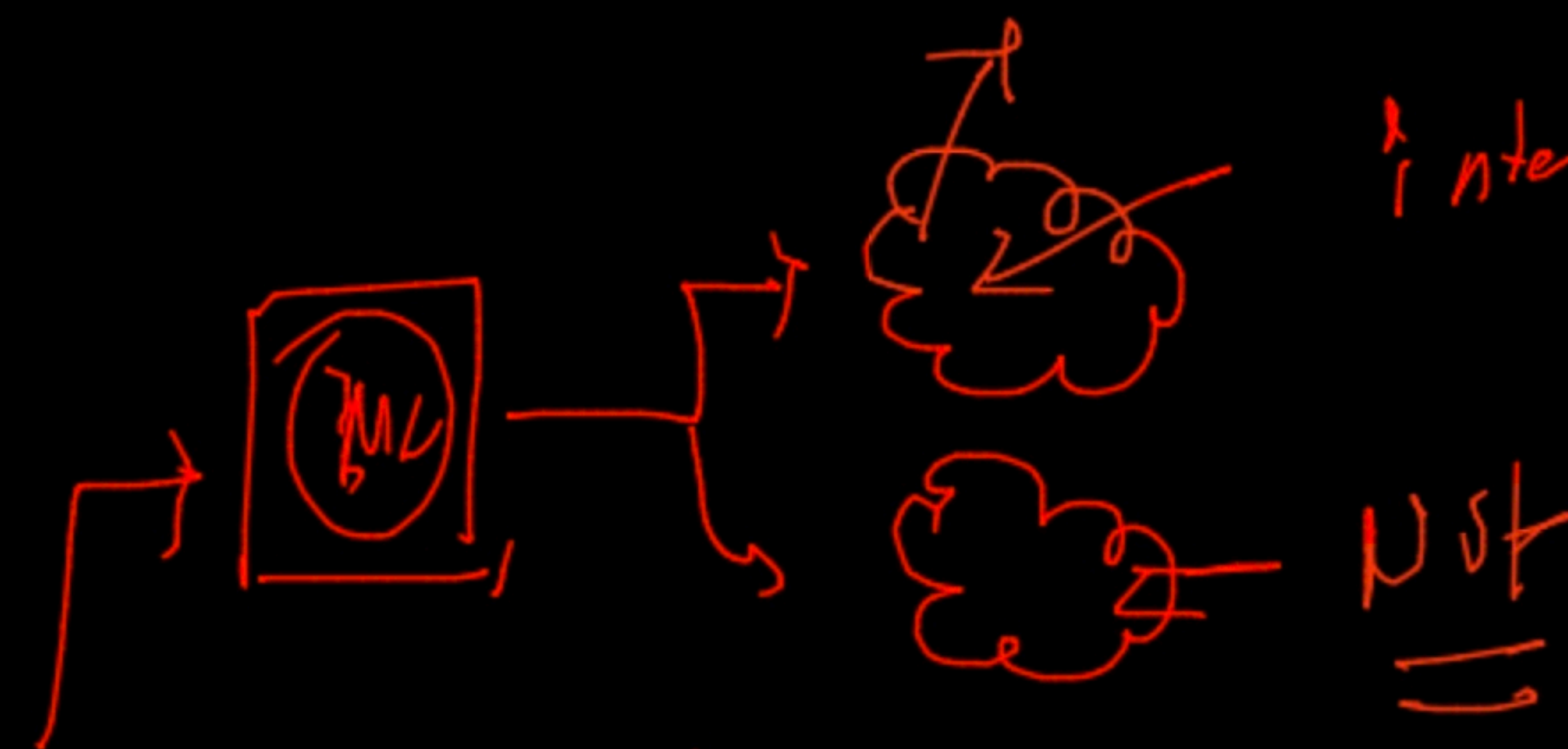
ex. 'Marketing team'

↑
Launch

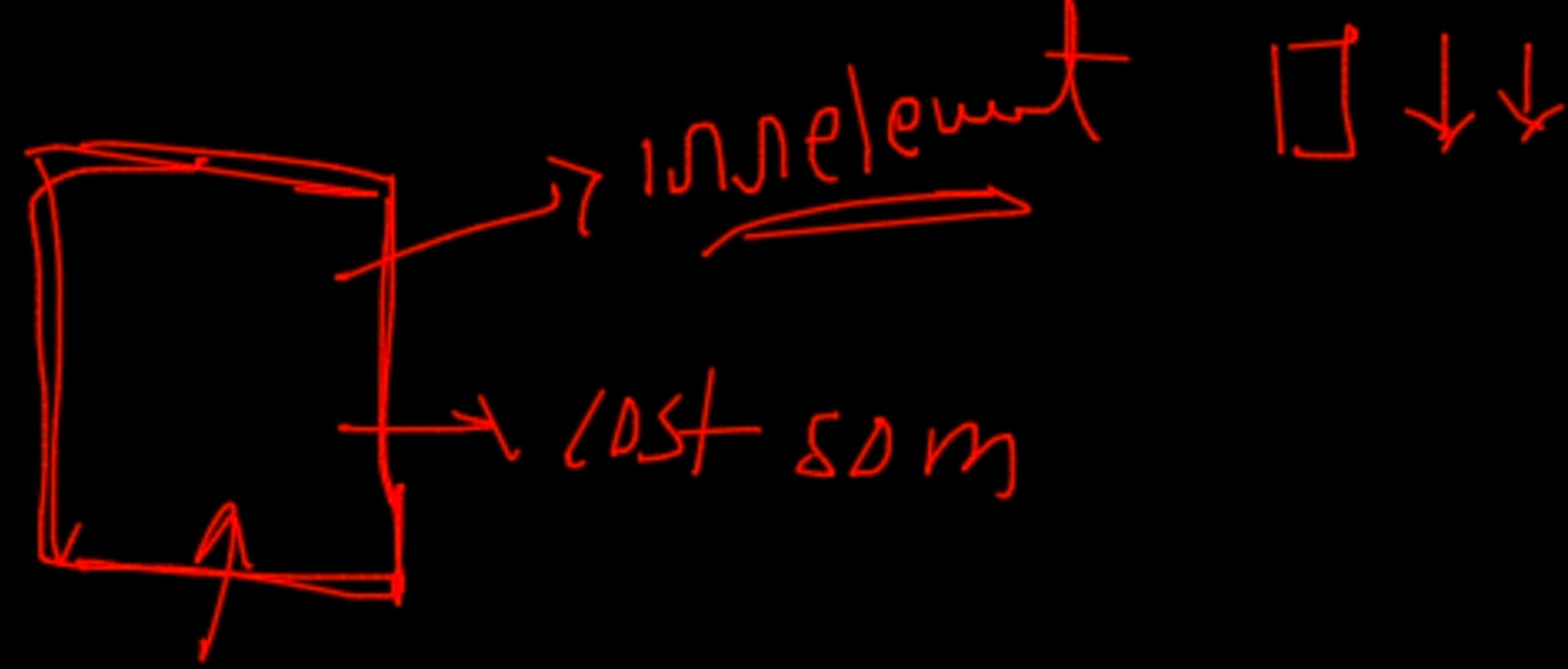
↑
'iPhone 7'

↑
advertisement

{ 'Instagram' }
face
}



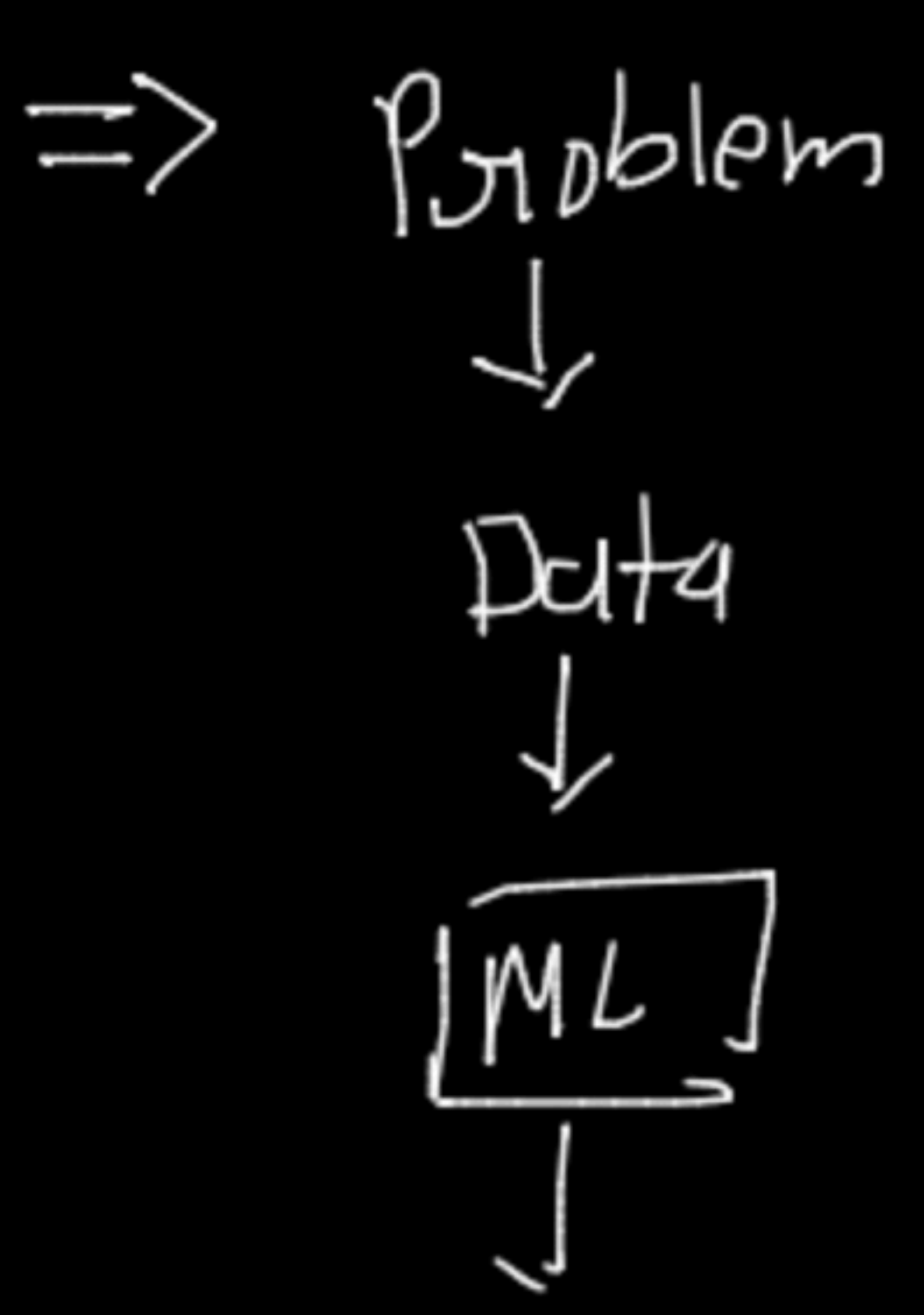
(tx, iphen, etc. ...)



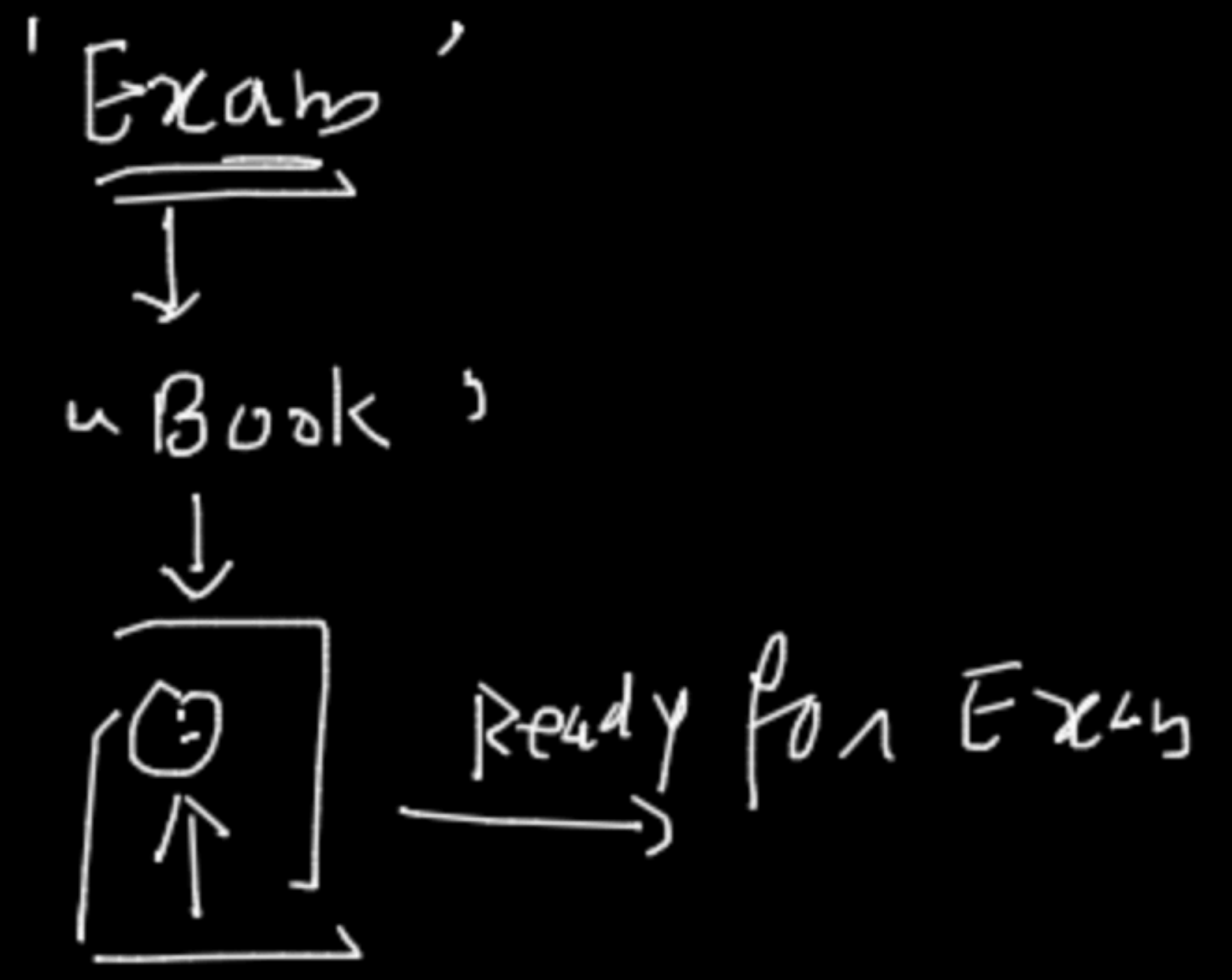
Model

(data) X
(tx iphen ...)

c1
c2
c3
c4
⋮
c10K

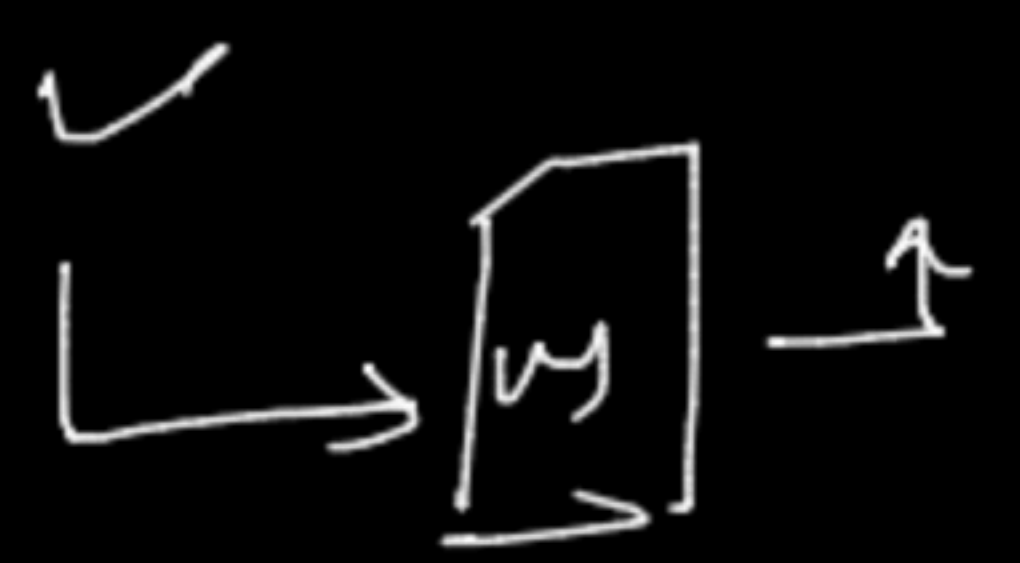


^Ready for
future g.

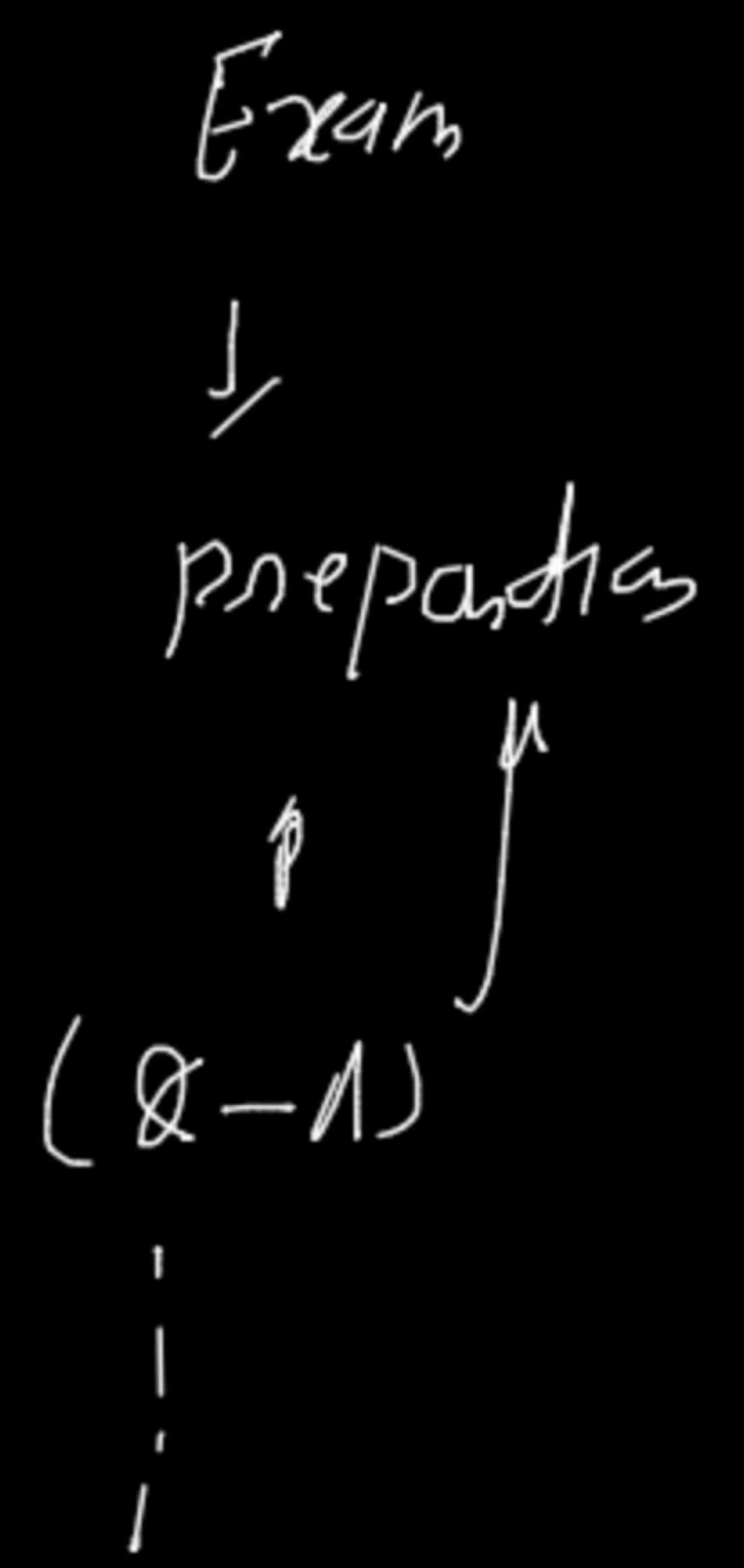
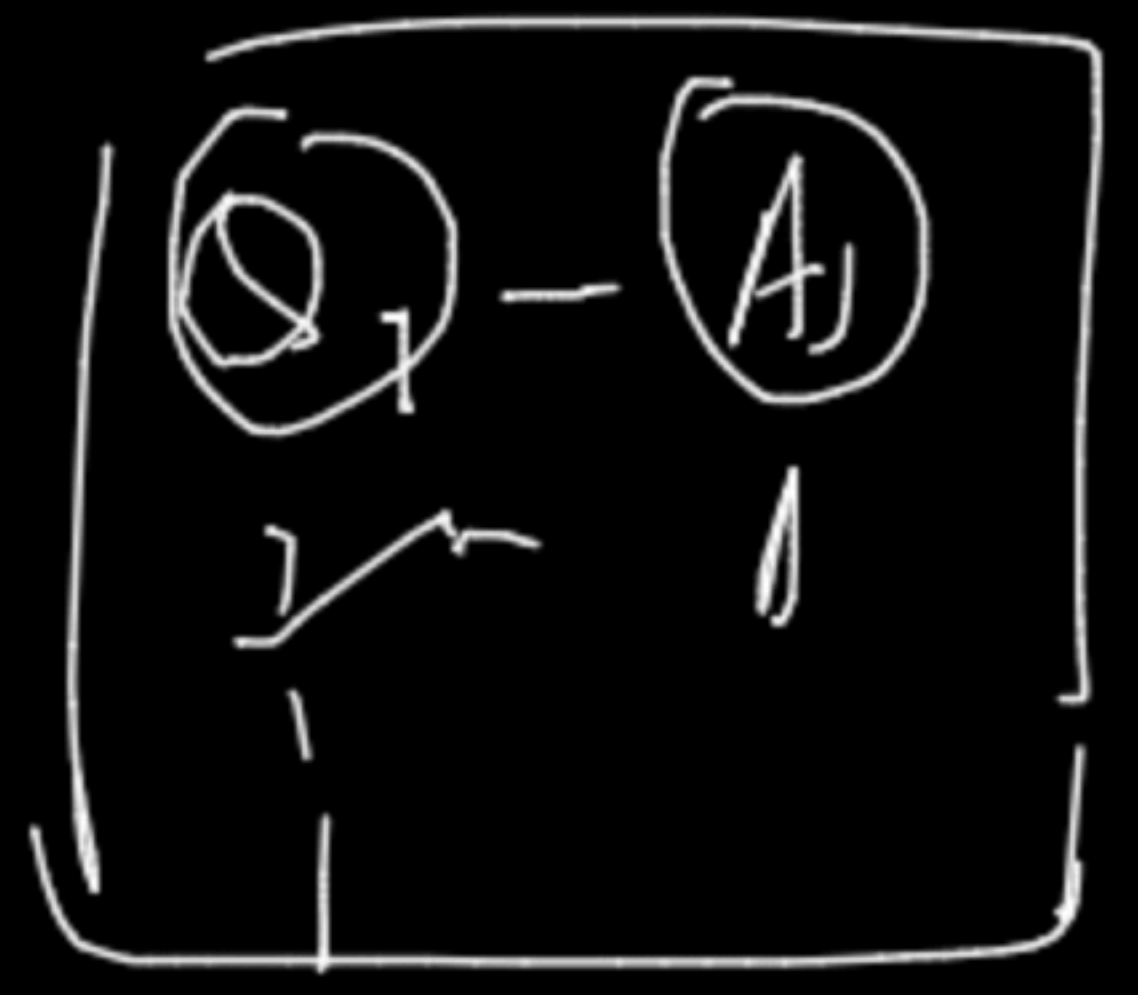
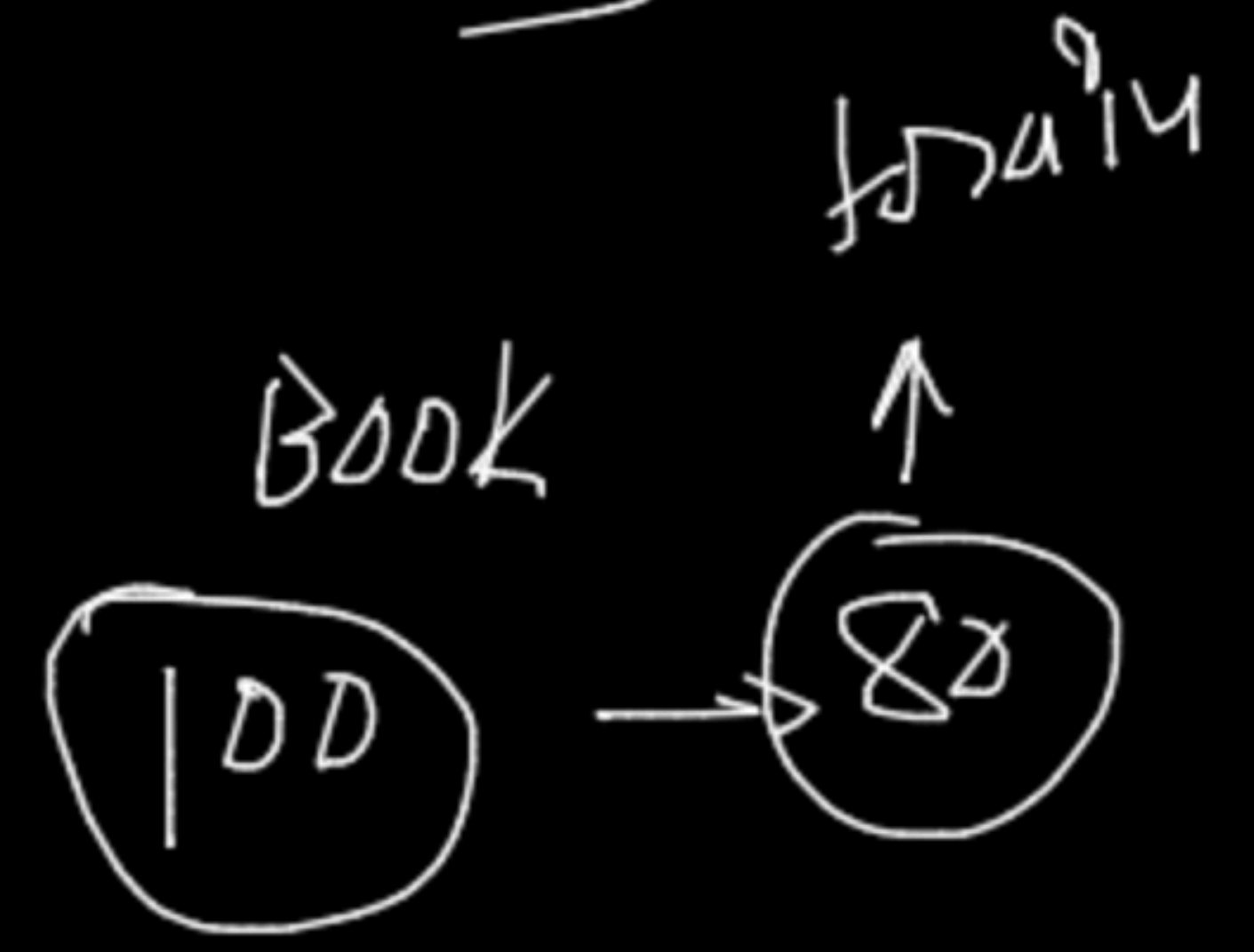


En

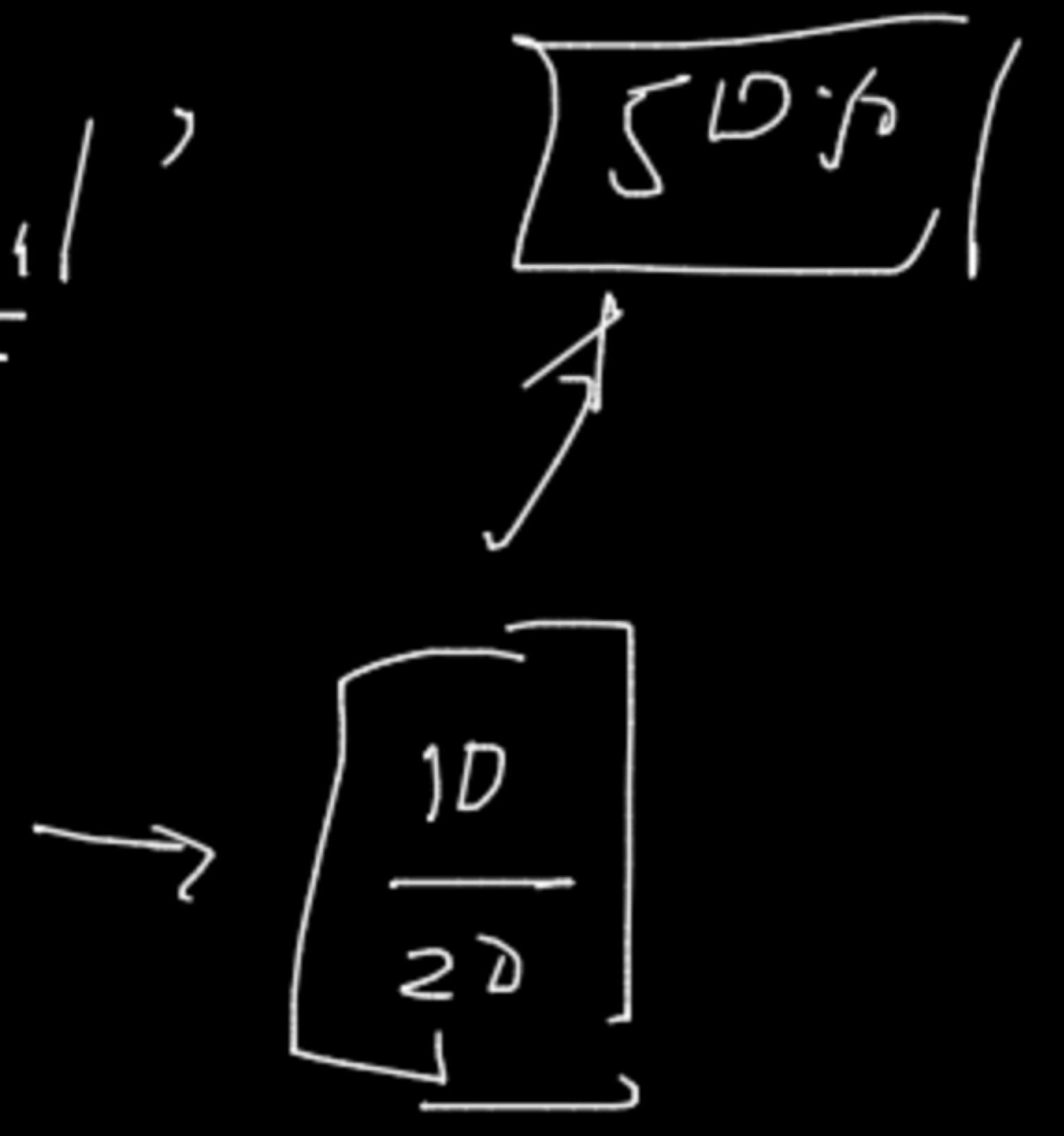
well?

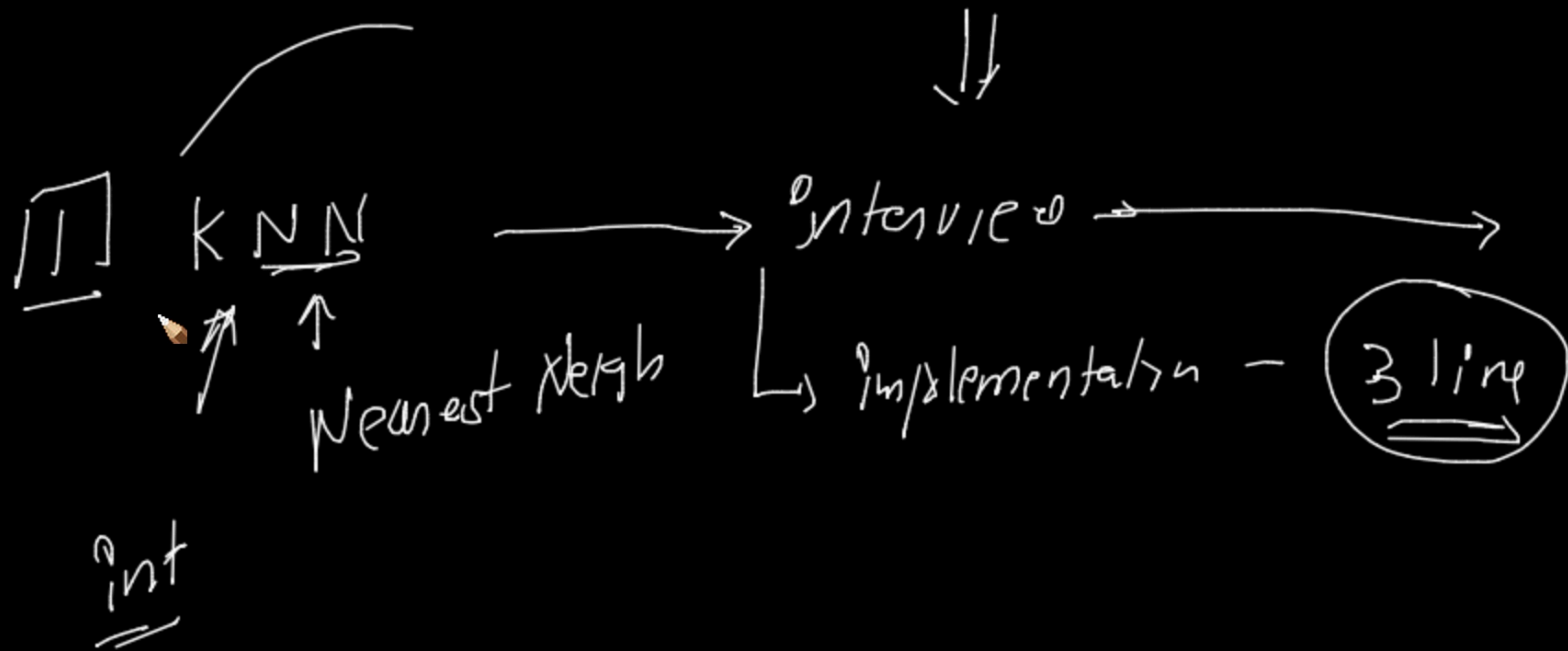
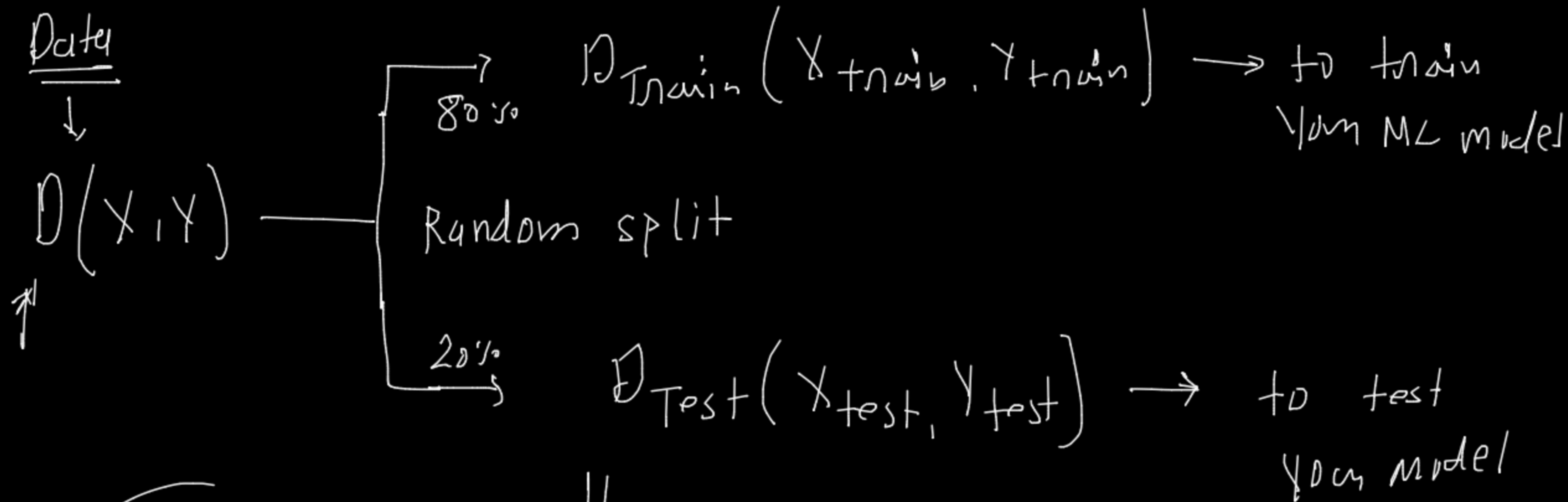


Here?



Final





$D(x, y)$

Historical

		x		y
	c-id	income	credit_score	status
	c1	✓ 17	750	Yes ✓
	c2	✓ 14	690	Yes ✓ Yes ✓
	c3	✓ 14	710	Yes ✓
	c4	✓ 14	655	Yes
	c5	✓ 3	455	No
	c6	✓ 7	450	No
	c7	✓ 4	555	No
	c8	✓ 2	400	No

Simple
KNOW

$D(x, y) \rightarrow \underline{D_{train}}(x_{train}, y_{train})$

17	750	Yes

$D(x, y) \rightarrow D_{test}(x_{test}, y_{test})$

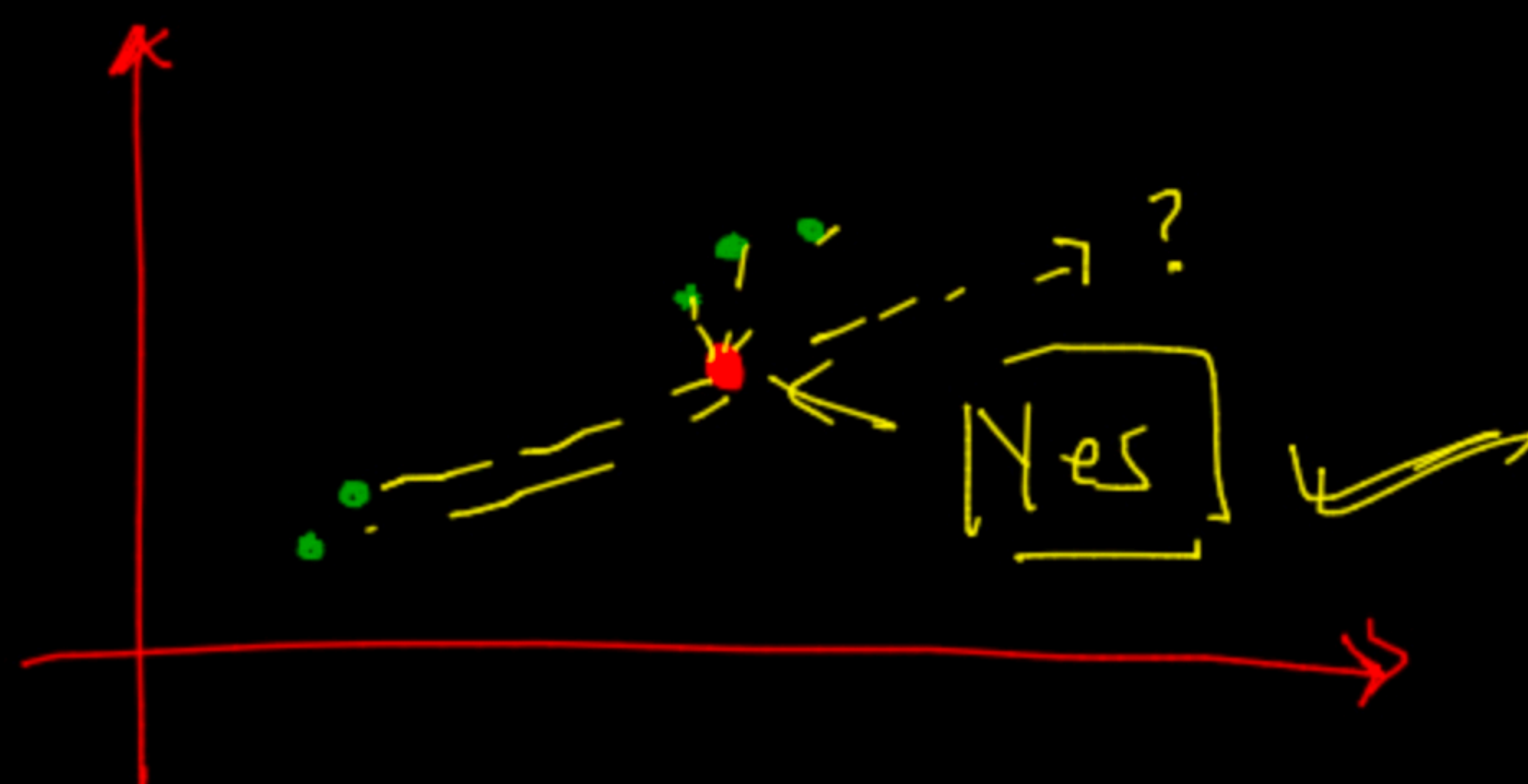
plot it

[2D]

~~K=3~~

KNN

Does not learn
anything → keep
data (trainings)
with him



→ at time of testing

↳ calculate the distance

↑
↳ sort →

Yes → 3 } ↑
No → 0 }

{
c1 → Yes
c3 → Yes
c4 → Yes